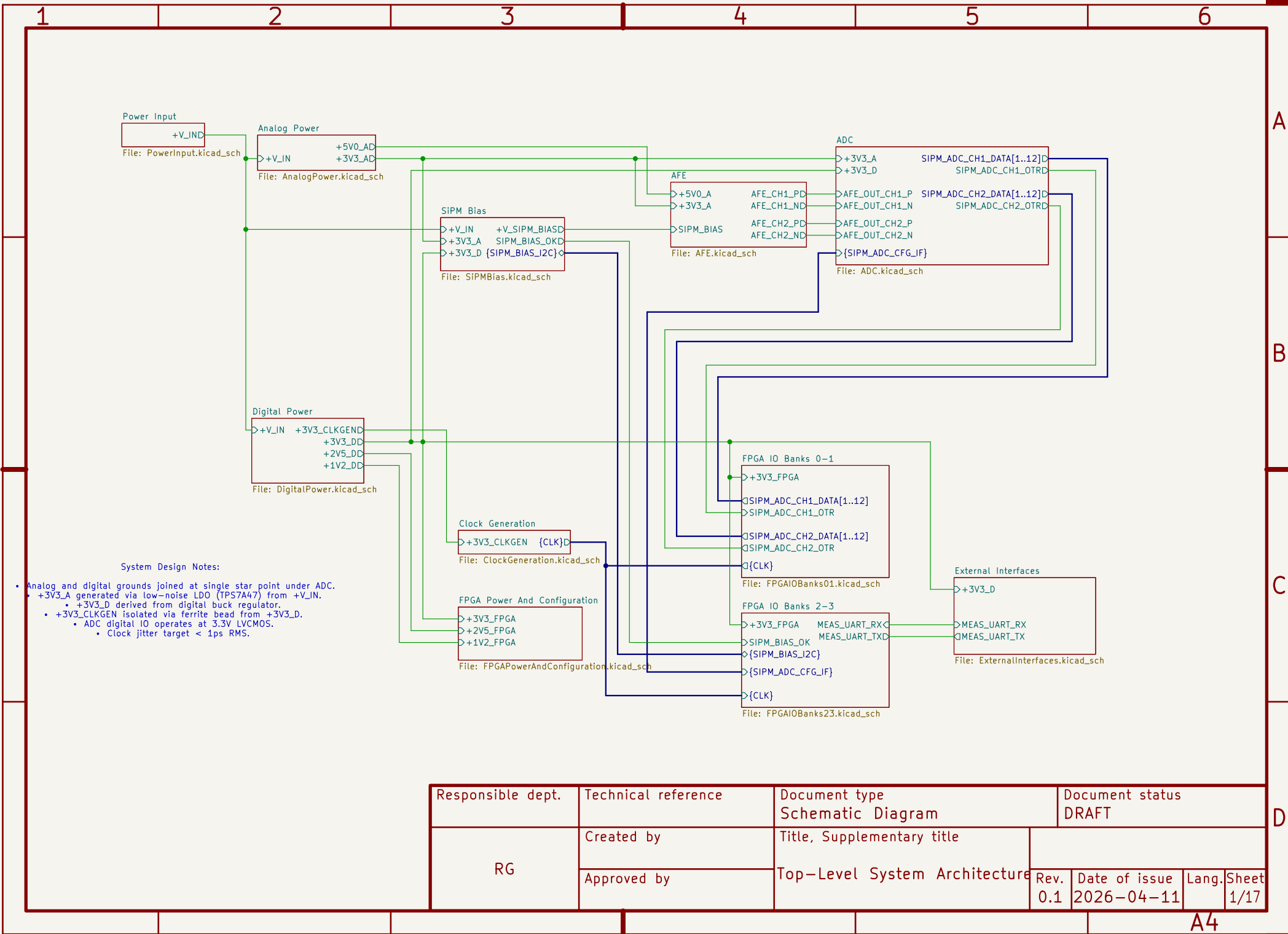
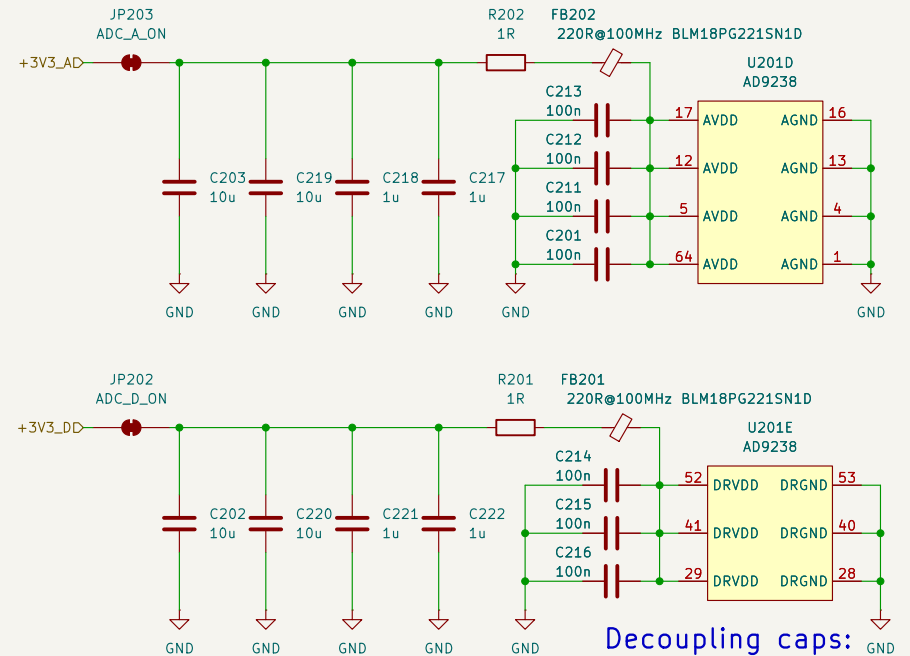
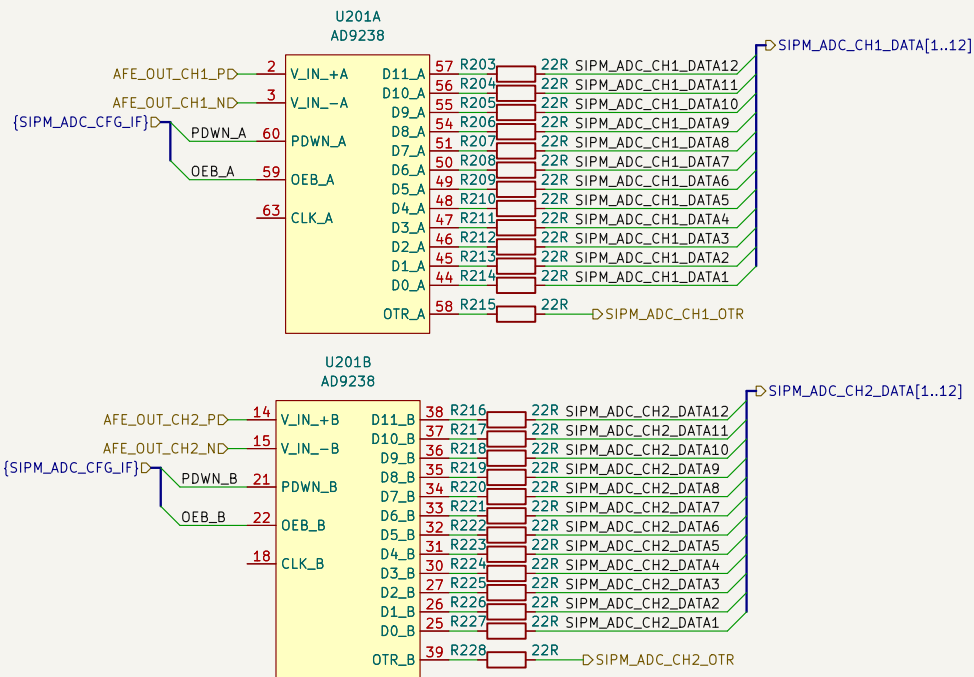
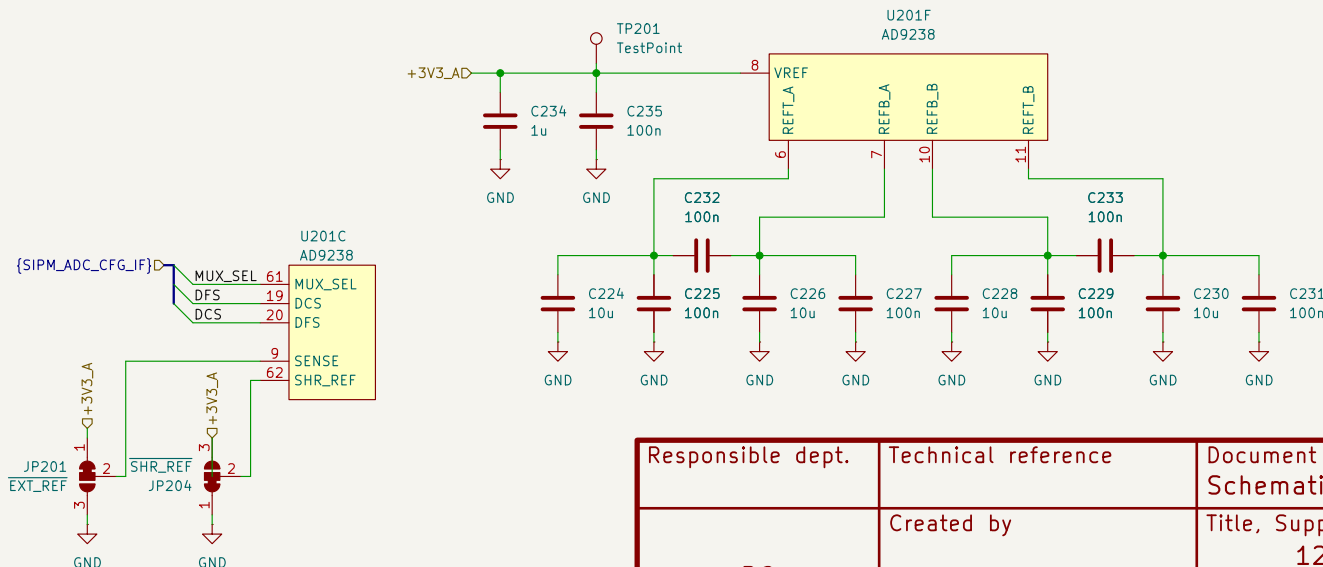






Decoupling caps:
100n per pin
1u+1u per rail
10u+10u per rail

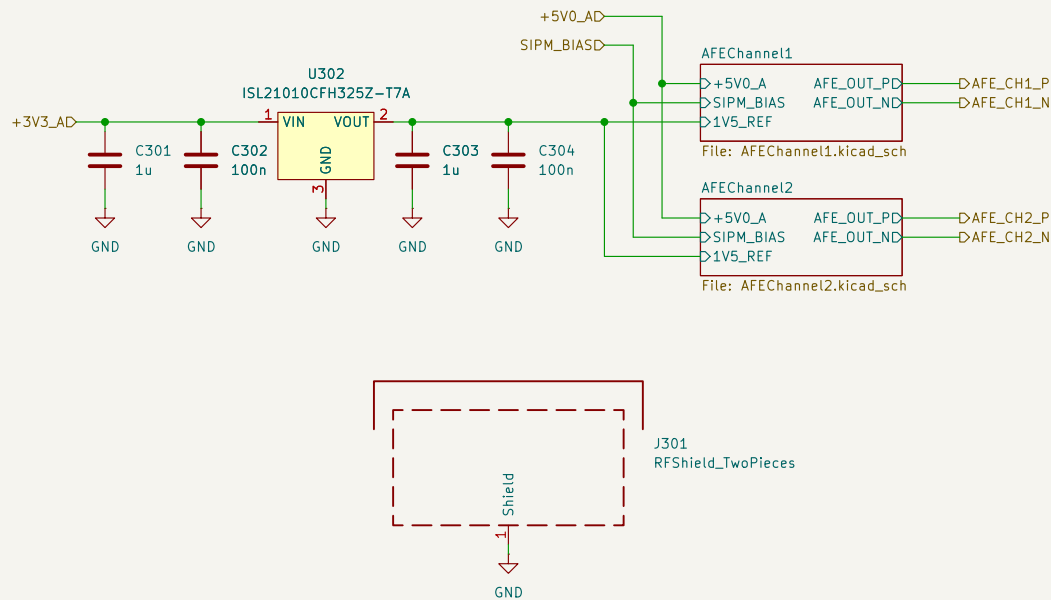
clocks from an FPGAs can have sharp edges that "ring" a



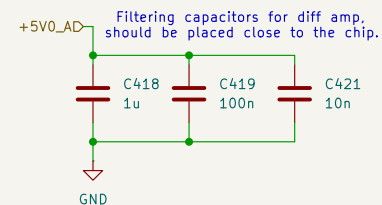
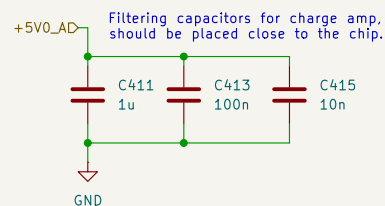
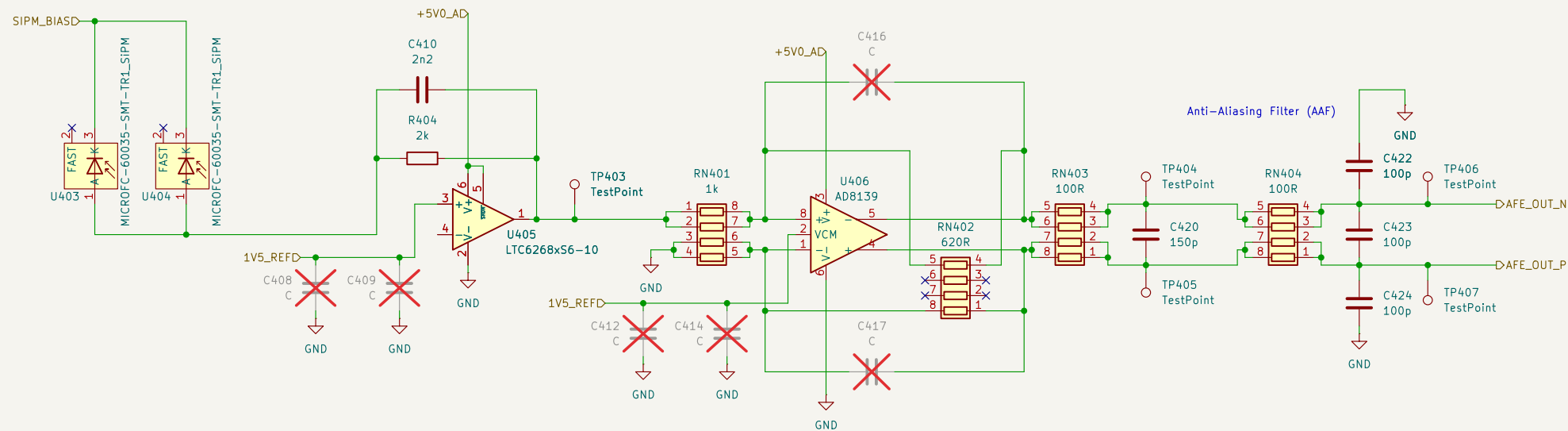
See AFE sheet for input conditioning

Pin Name	Description
VREF	ADC reference voltage (input/output)
SENSE	Select reference mode, AVDD for external
SHARED_REF	High = shared reference, Low = independent
MUX_SEL	Low = multiplex A/B data, High = normal mode
DCS	High = enable duty cycle stabilizer
DFS	Low = offset binary, High = 2's complement
OEB_x	Low = outputs enabled, High = Hi-Z

Responsible dept.	Technical reference	Document type	Document status
RG	Created by	Schematic Diagram	DRAFT
	Approved by	Title, Supplementary title	
		12-bit 65MSPS Analog-to-Digital Converter	
		Rev. 0.1	Date of issue 2026-04-11
		Lang.	Sheet 2/17

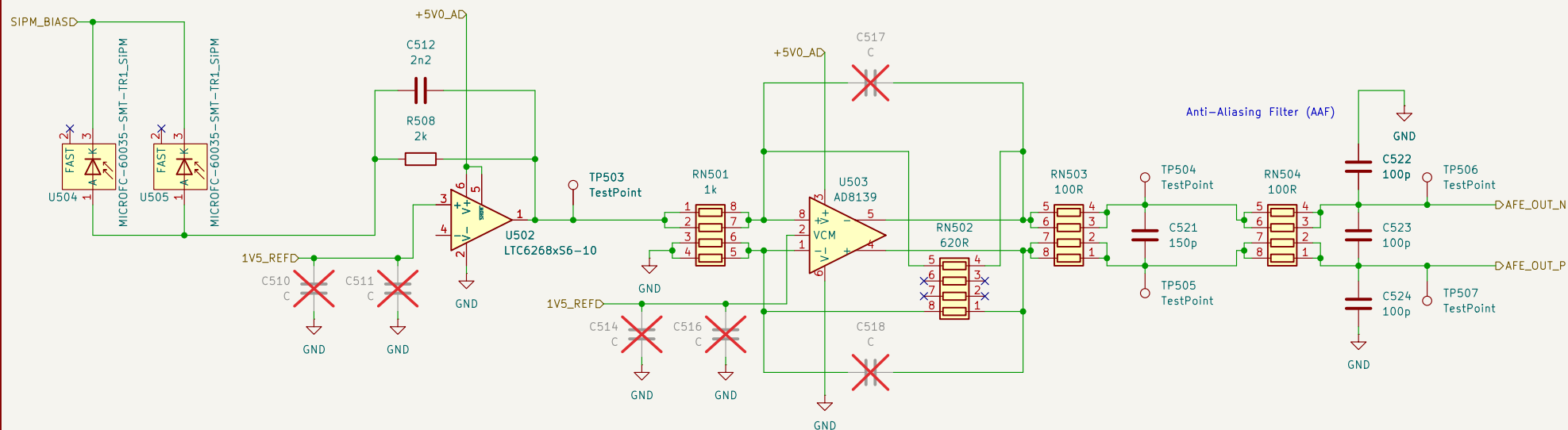


Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by	TIA & Differential ADC Driver	Rev. 0.1 Date of issue 2026-04-11 Lang. Sheet 3/17



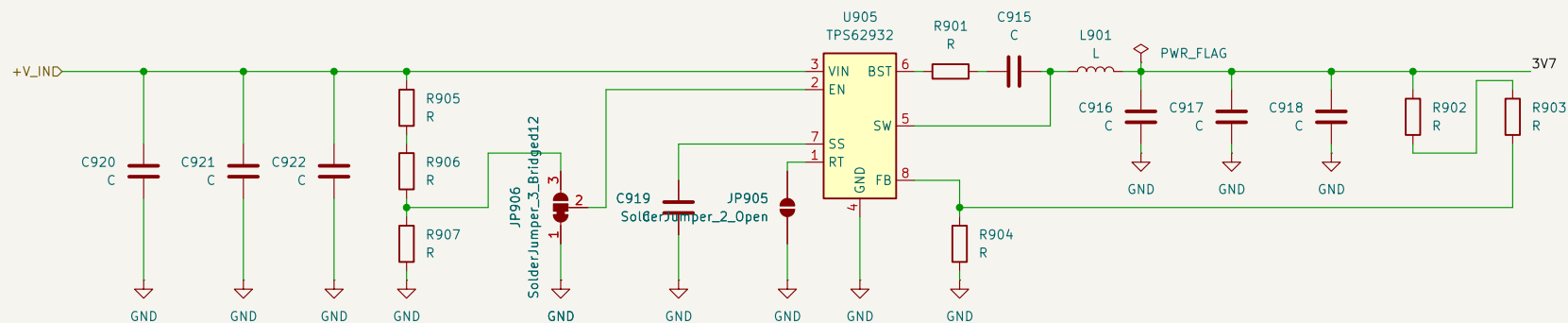
All capacititors (except for voltage filtering) are COG / NP0.
All resistors are 1% or better.

Responsible dept.	Technical reference	Document type		Document status			
	Created by	Title, Supplementary title					
	Approved by						
		Rev.	Date of issue	Lang.	Sheet		
					4/17		



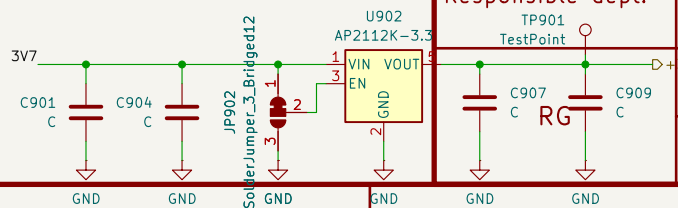
All capcitors (except for voltage filtering) are COG / NP0.
All resistors are 1% or better.

Responsible dept.	Technical reference	Document type	Document status			
	Created by	Title, Supplementary title				
	Approved by		Rev.	Date of issue	Lang.	Sheet
						5/17

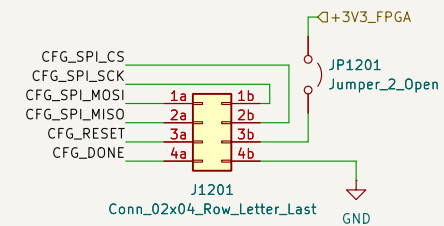
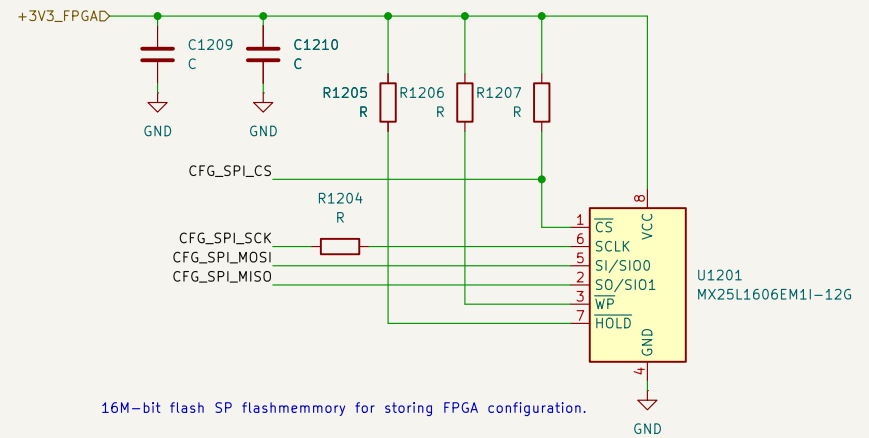
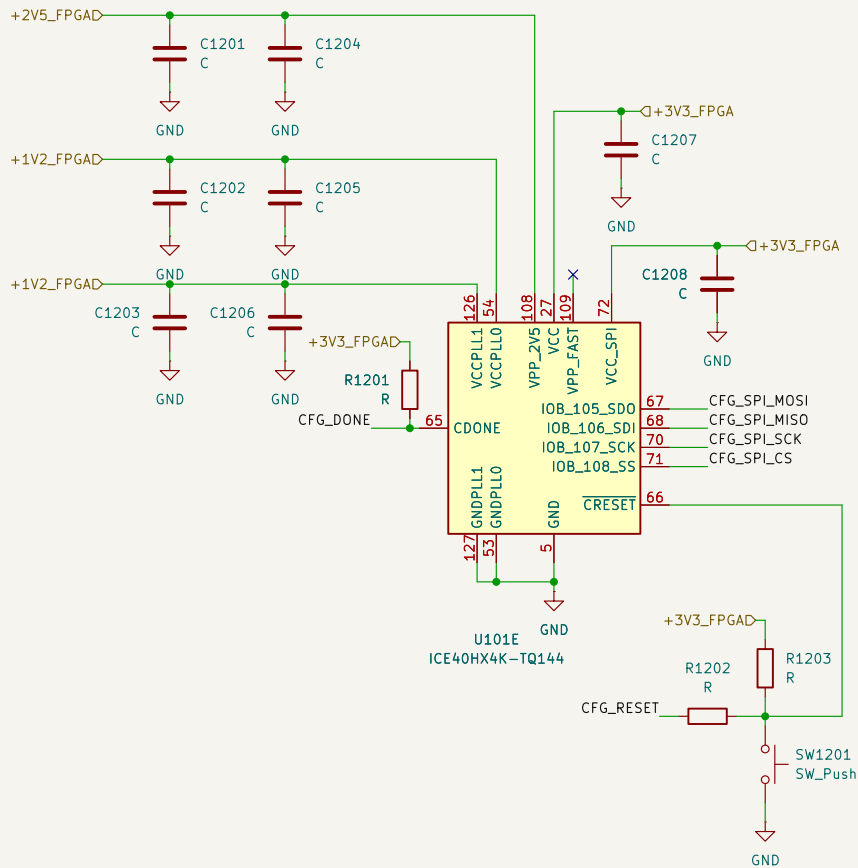


FIX VALUES!!!!

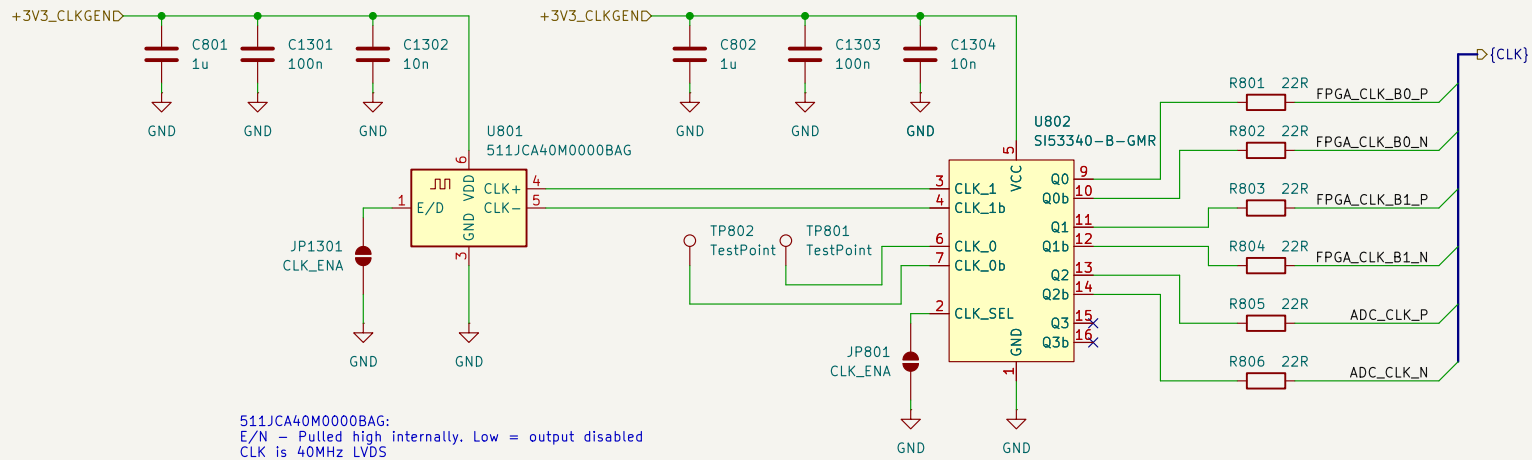
Label	Direction	Description
VCC	Input	Raw input voltage (6V)
VCC_SIPM	Output	30V for analog frontend bias
VCC_3V3	Output	3.3V for FPGA I/O, flash, ADC digital side
VCC_1V2	Output	1.2V for FPGA core
VCC_5V	Output	5V for ADC
VCC_2V5	Output	2V5 for amplifier
VEE_2V5	Output	-2V5 for amplifier
GND	Bidirectional	Ground



Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
	Created by	Title, Supplementary title	
	Approved by		
		Digital Power Regulators (3.3V, 1.2V)	Rev. 0.1
			Date of issue 2026-04-11
			Lang.
			Sheet 6/17



Responsible dept.	Technical reference		Document type		Document status			
			Schematic Diagram		DRAFT			
RG	Created by		Title, Supplementary title					
	Approved by		FPGA Configuration Memory – External 16Mb SPI Boot Flash		Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 7/17

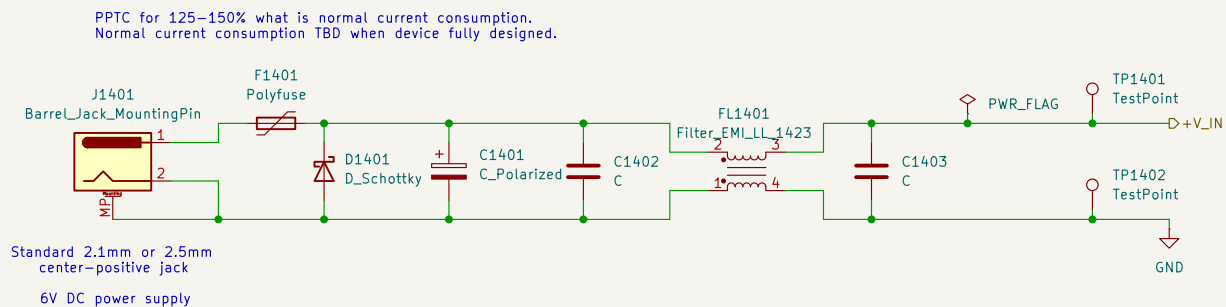


511JCA40M0000BAG:
E/N – Pulled high internally. Low = output disabled
CLK is 40MHz LVDS

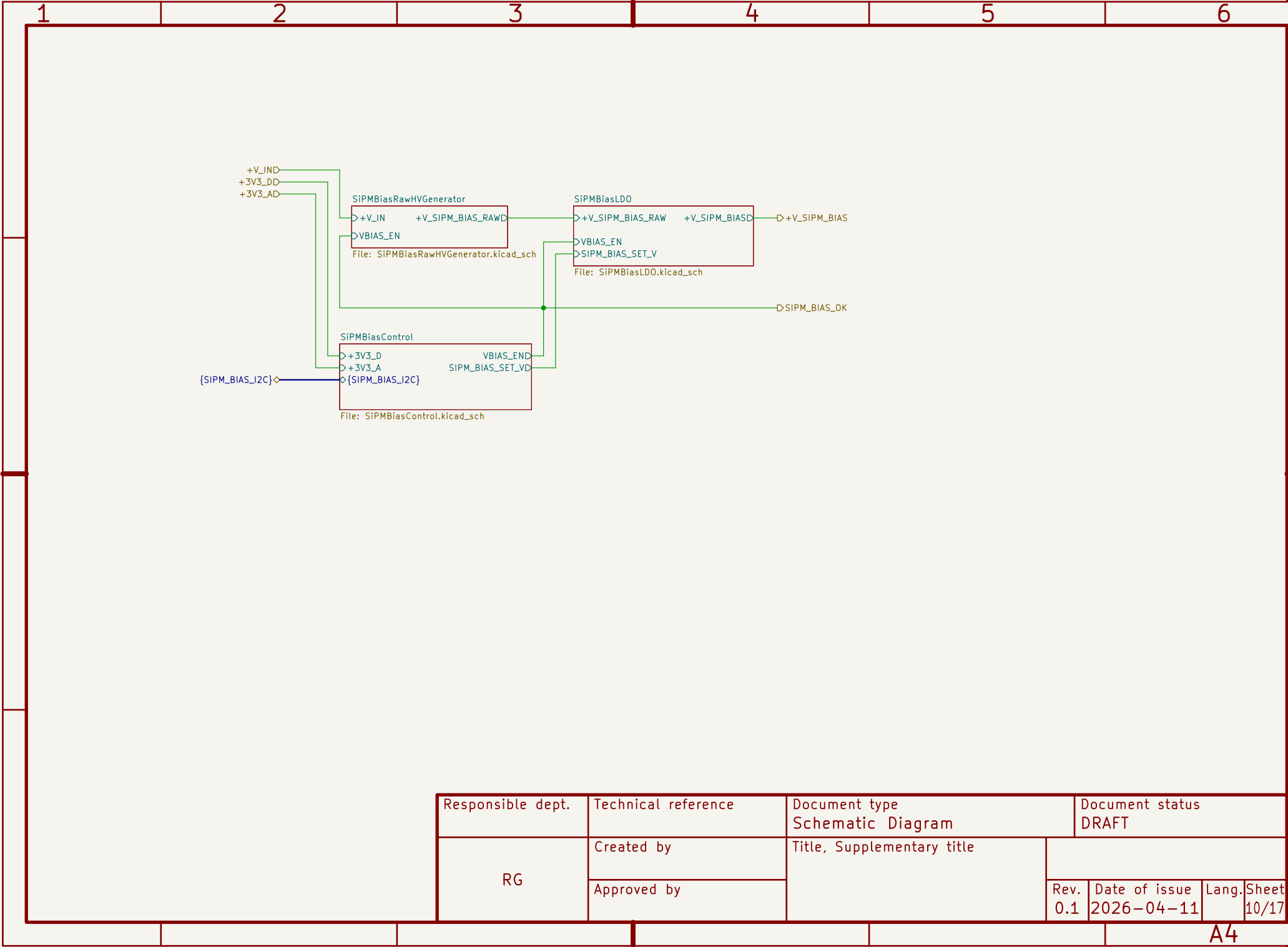
SI53340-B-GMR: When CLK_SEL is high,
CLK1 is selected. When CLK_SEL is low,
CLK0 is selected. CLK_SEL contains
an internal pull-down resistor

TODD for PCB:
Impedance control
Skew budget

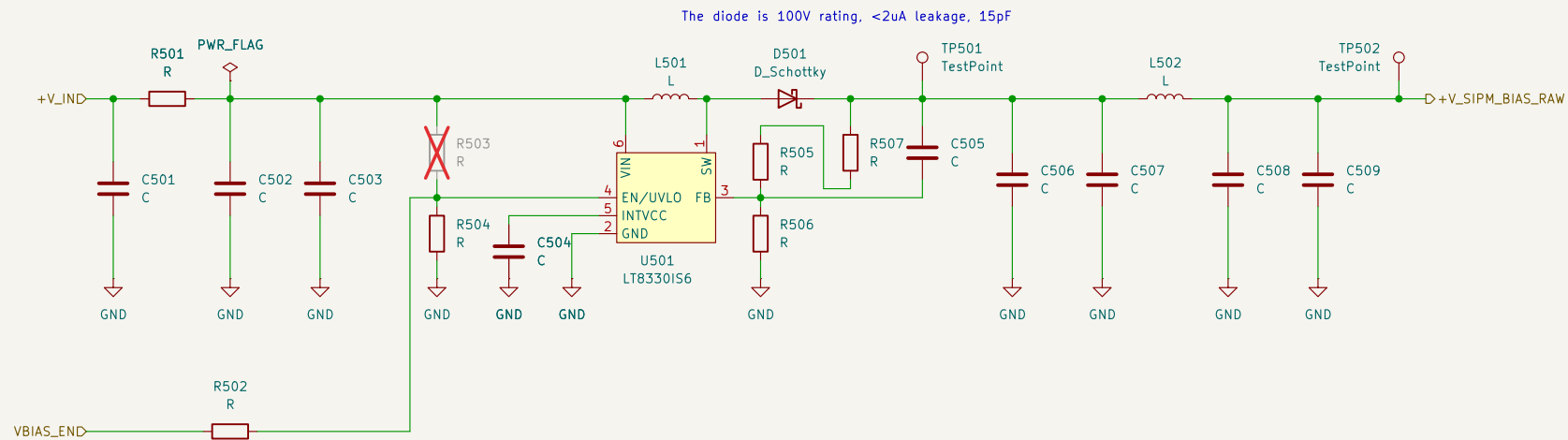
Responsible dept.	Technical reference		Document type		Document status			
			Schematic Diagram		DRAFT			
RG	Created by		Title, Supplementary title					
	Approved by		System Clock – 65 MHz Oscillator and Dual Output Buffer		Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 8/17



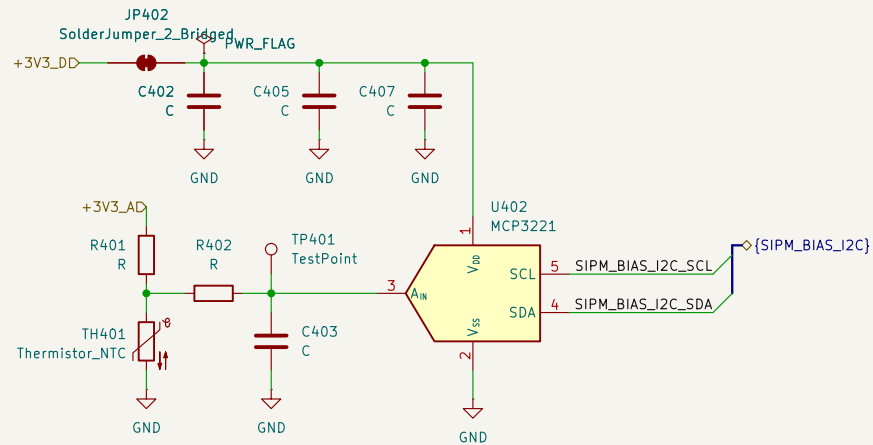
Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by	Barrel Jack Input and Preliminary Protection Interface	
	Rev.	Date of issue	Lang. Sheet
	0.1	2026-04-11	9/17



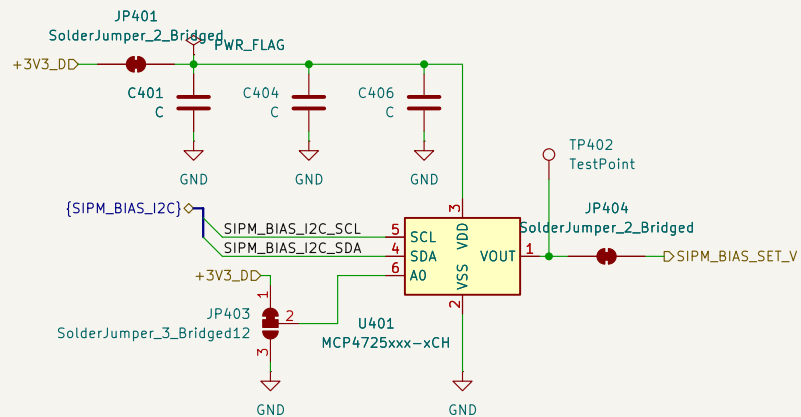
Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT			
	Created by	Title, Supplementary title					
	Approved by			Rev.	Date of issue	Lang.	Sheet
RG				0.1	2026-04-11		10/17



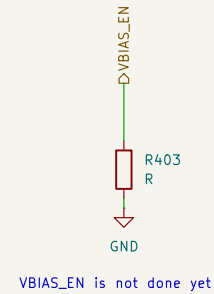
Responsible dept.	Technical reference	Document type Schematic Diagram	Document status DRAFT			
RG	Created by	Title, Supplementary title High-Voltage SiPM Bias Generator		Rev. 0.1 Date of issue 2026-04-11 Lang. Sheet 11/17		
	Approved by					



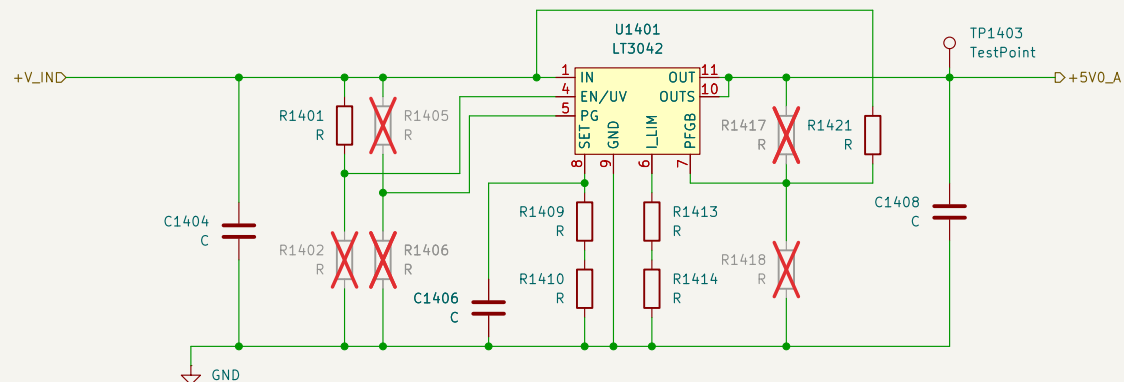
MCP3221A5T-E/OT: 12-bit I2C ADC with a fixed I2C address.



MCP4725A0T-E/CH is a 12-bit I2C DAC with an address that can be modified via the A0 pin.

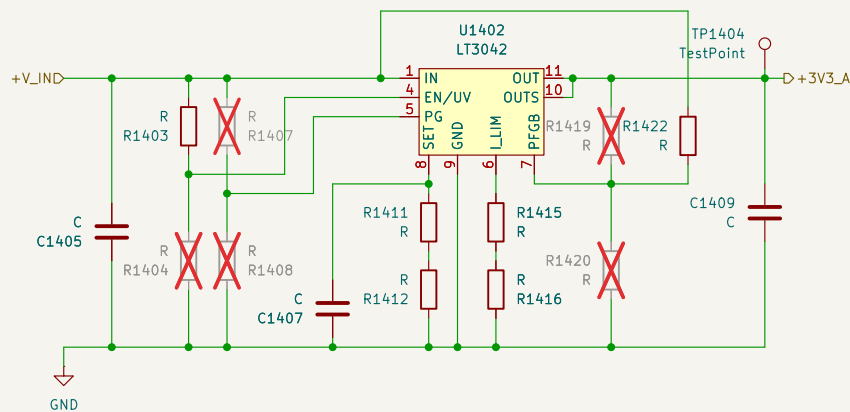


Responsible dept.	Technical reference	Document type	Document status
		Schematic Diagram	DRAFT
RG	Created by	Title, Supplementary title	
	Approved by	SIPM Bias Control and Temperature Compensation	
	REV. 0.1	Date of issue 2026-04-11	Lang. Sheet 12/17



OUTS is a sensor pin. Should be connected close to the IC that is supplied by this stabilizer on the PCB.

HIGH QUALITY CAPS FOR SET PIN!!!



OUTS is a sensor pin. Should be connected close to the IC that is supplied by this stabilizer on the PCB.

HIGH QUALITY CAPS FOR SET PIN!!!

LT3042 calculation:

$$R_{set} = \frac{\text{abs}(V_{out})}{100\mu A}$$

$$I_{limit} = \frac{125\text{mA} \cdot k\Omega}{R_{lim}}$$

For:

$$\text{for } 60 \text{ mA: } R_{lim} = \frac{125}{60} = 2.08\text{k}$$

$$\text{for } 70 \text{ mA: } R_{lim} = \frac{125}{70} = 1.79\text{k}$$

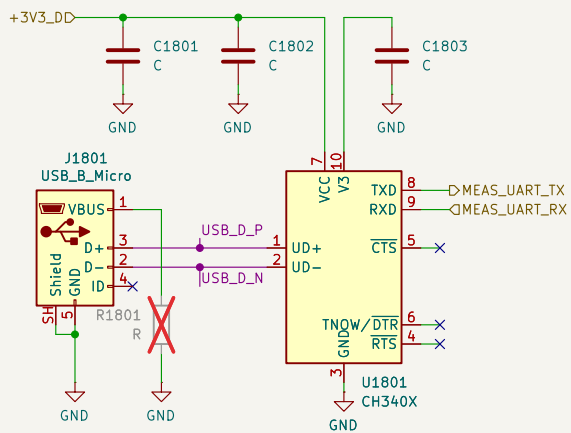
$$\text{for } 100 \text{ mA: } R_{lim} = \frac{125}{100} = 1.24\text{k}$$

PG not used -> leave PG floating

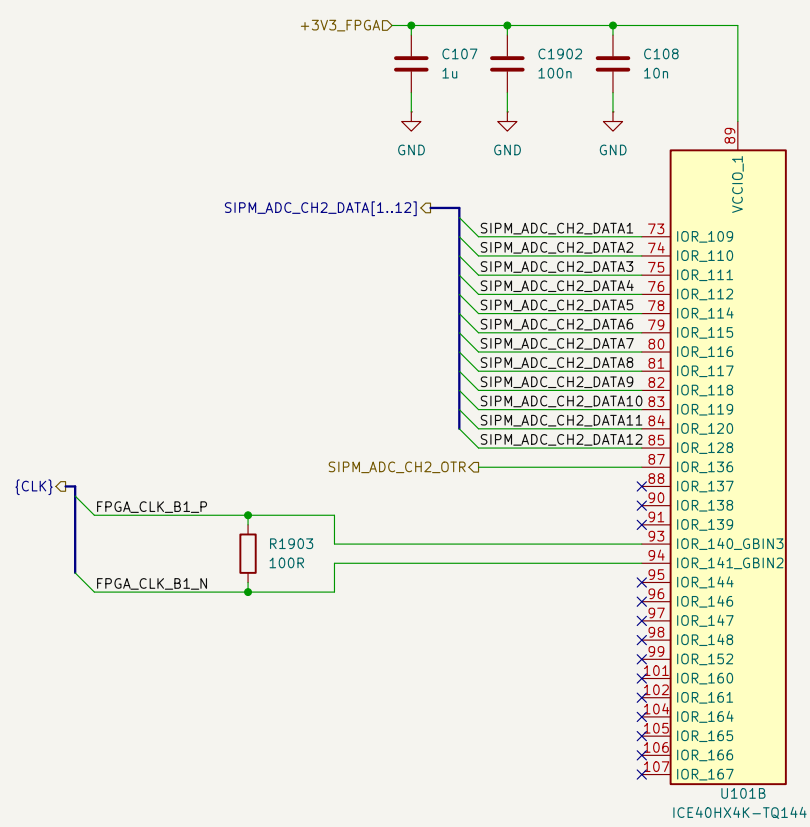
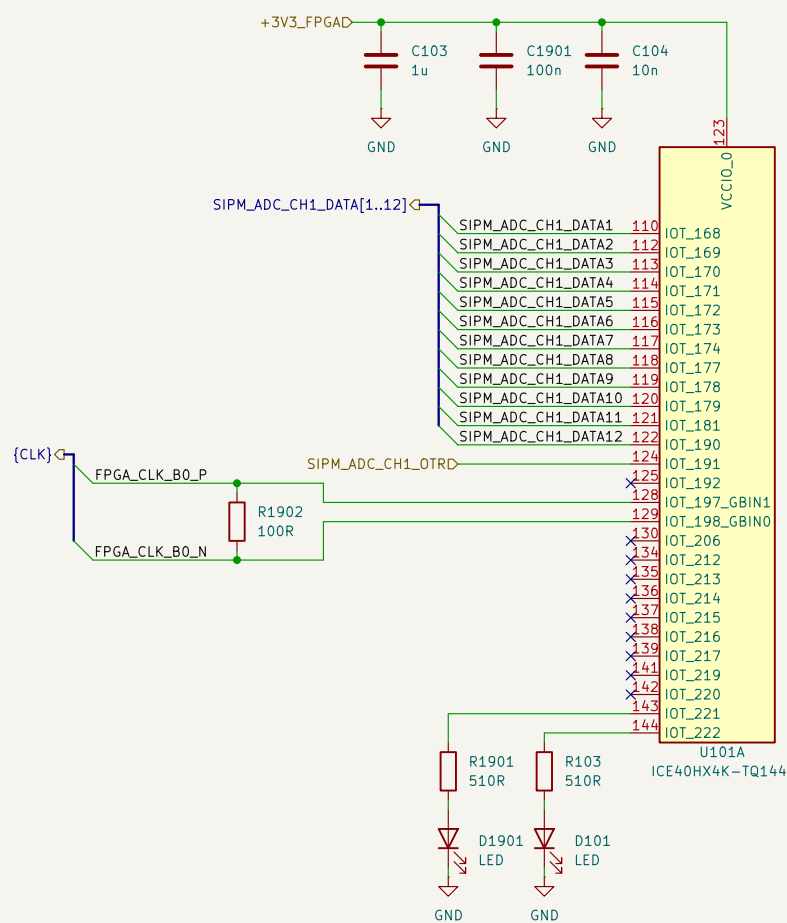
PGFB not used -> tie PGFB to IN

EN/UV tied to IN for always-on operation

Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT			
RG	Created by	Title, Supplementary title					
	Approved by	Analog Power		Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 14/17



Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT		
RG	Created by	Title, Supplementary title External Interfaces				
	Approved by			Rev. 0.1	Date of issue 2026-04-11	Lang.



Responsible dept.	Technical reference	Document type Schematic Diagram		Document status DRAFT			
RG	Created by	Title, Supplementary title					
	Approved by		Rev. 0.1	Date of issue 2026-04-11	Lang.	Sheet 19/17	